

sensitivity is small (negative log) over a given range of frequencies, there is only a finite frequency range over which we can spread the required sensitivity gain or “positive log” area. Again, unstable poles make the situation worse and the effect increases if the poles are located far from the unit circle. We can illustrate the implications of Eq. (7.50) by two simple examples. Consider a stable generic model of a chemical process and an unstable model of a magnetic levitation system. The two transfer functions are

$$G_1(s) = \frac{1}{(s+1)^2} \quad \text{chemical process,}$$

$$G_2(s) = \frac{1}{s^2 - 1} \quad \text{magnetic levitation.}$$

In each case we will include a zero-order hold and sample with $T = 0.2$ and in each case the controller transfer function is $D = 15 \frac{z-0.7}{z+0.5}$, selected to give approximately the same rise time and bandwidth. The step responses are shown in Fig. 7.31(a) and the sensitivity plots are shown in Fig. 7.31(b). Notice the substantially larger value of the sensitivity for the unstable plant compared to that of the stable one. The vector gain margin for G_1 is 5.4 while that for G_2 is 2.25, less than half as much. To plot the sensitivity magnitude, it is necessary to obtain a system having the transfer function of S . This can be done using the feedback function if a representation of the open loop system DG is given as, for example, the open loop discrete system sysold. The expression syscl = feedback(sys1,sys2) generates the loop with forward system sys1 and feedback system sys2. For sensitivity, DG is the feedback system, and we need to construct the dummy gain of 1 for the forward system. This can be done with the statement sysf = ss(0,0,0,1,Ts). Finally, the sensitivity is given by sens = feedback(sysf,sysold). The plot is given by the statements

```
[mag,ph,w]=bode(sens);
semilogy(w,mag)
grid.
```

weighted integral of the discrete sensitivity function

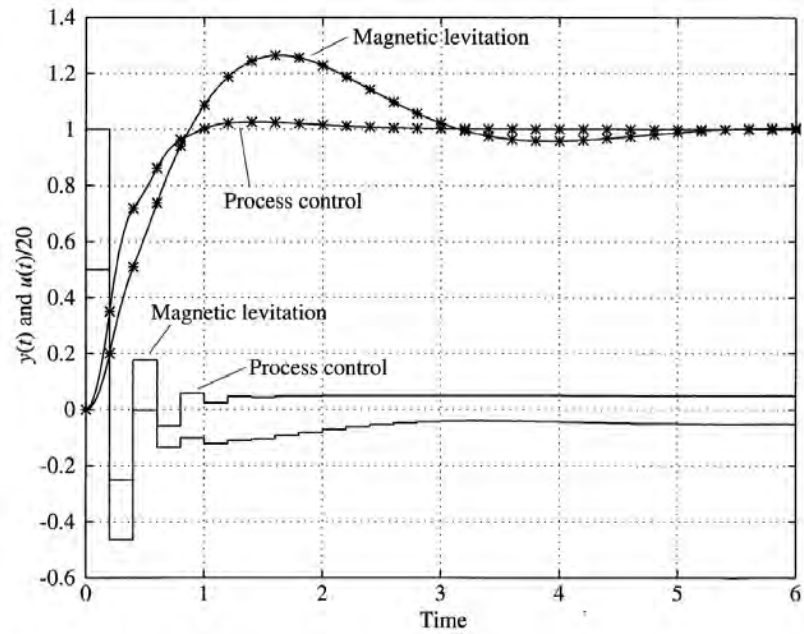
The weighted integral of the discrete sensitivity function is similar to that of the continuous case. We assume again that the system has n_p poles outside the unit circle at $\alpha_i = r_i e^{j\phi_i}$, $r_i > 1$, and conjugate $\bar{\alpha}_i$, and also has a zero outside the unit circle at $\beta_o = r_o e^{j\phi_o}$, $r_o > 1$. Then

$$\int_{-\pi}^{\pi} \ln(|S(e^{j\phi})|) \frac{r_o^2 - 1}{r_o^2 - 2r_o \cos(\phi - \phi_o) + 1} d\phi = 2\pi \sum_{i=1}^{n_p} \ln \left| \frac{1 - \bar{\alpha}_i \beta_o}{\beta_o - \alpha_i} \right|. \quad (7.51)$$

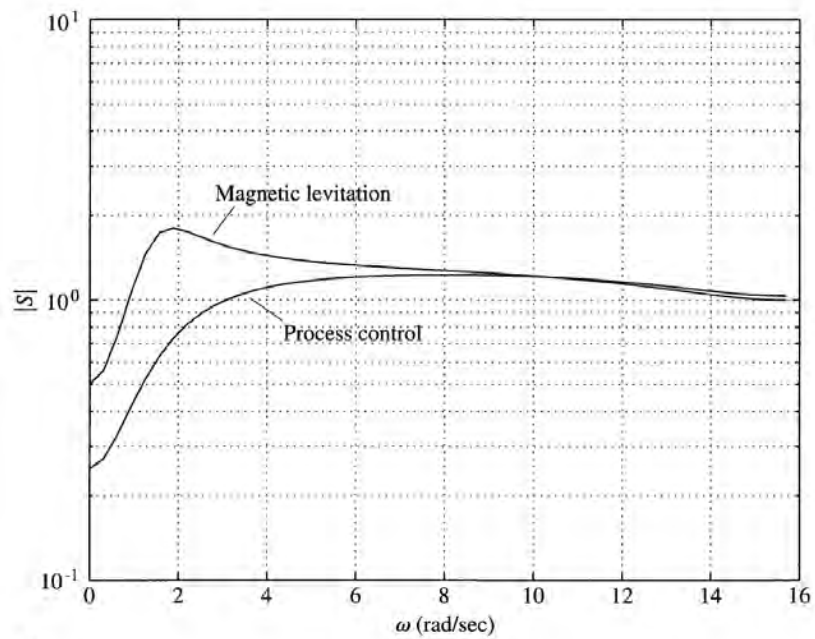
The main consequence of this constraint is that it expresses a limitation imposed by the non-minimum phase zero on the sensitivity function. The constraint is especially severe if there is a non-minimum phase zero near an unstable pole ($\beta_o \approx \alpha_i$).

Figure 7.31

Comparisons of a stable process control with an unstable magnetic levitation: (a) step responses (b) sensitivity plots



(a)



(b)

7.4.3 Low Frequency Gains and Error Coefficients

The steady-state error constants for polynomial inputs for discrete systems were established in Section 7.2 and are given by

$$K_p = \lim_{z \rightarrow 1} D(z)G(z)$$

for a Type 0 system, and by

$$K_v = \lim_{z \rightarrow 1} \frac{(z-1)D(z)G(z)}{Tz}, \quad (7.52)$$

for a Type 1 system. In the frequency domain, for a Type 0 system, the procedure is identical to the continuous case. Since $z = e^{j\omega T}$, $z \rightarrow 1$ implies that $\omega T \rightarrow 0$, and the magnitude frequency-response plot will show a constant value on the low-frequency asymptote which is equal to K_p . For a Type 1 system, the procedure is again identical to the continuous case in that the magnitude of $D(z)G(z)$ at $\omega = 1$ on the low-frequency asymptote is equal to K_v . This can be seen from Eq. (7.52) if we note that for $\omega T \rightarrow 0$, $e^{j\omega T} \cong 1 + j\omega T$. Therefore

$$\lim_{z \rightarrow 1} \frac{(z-1)}{Tz} = \lim_{j\omega \rightarrow 0} \omega,$$

thus establishing the fact that evaluation of the low-frequency asymptote of $D(z)G(z)$ at $\omega = 1$ yields K_v . This fact is most easily used if the frequency-response magnitude is plotted versus ω in units of rad/sec so that $\omega = 1$ rad/sec is readily found. If the magnitude is plotted versus ω in units of Hz or versus ωT , one would need to perform a calculation to find the $\omega = 1$ rad/sec point. However, the error constants could be calculated directly with good software tools; therefore the issues in their calculation are of passing interest only. But no matter how the constants are found, *the fact remains for discrete and continuous frequency response alike, the higher the magnitude curve at low frequency, the lower the steady-state errors.*

◆ Example 7.14 Finding Velocity Constant on a Bode Plot

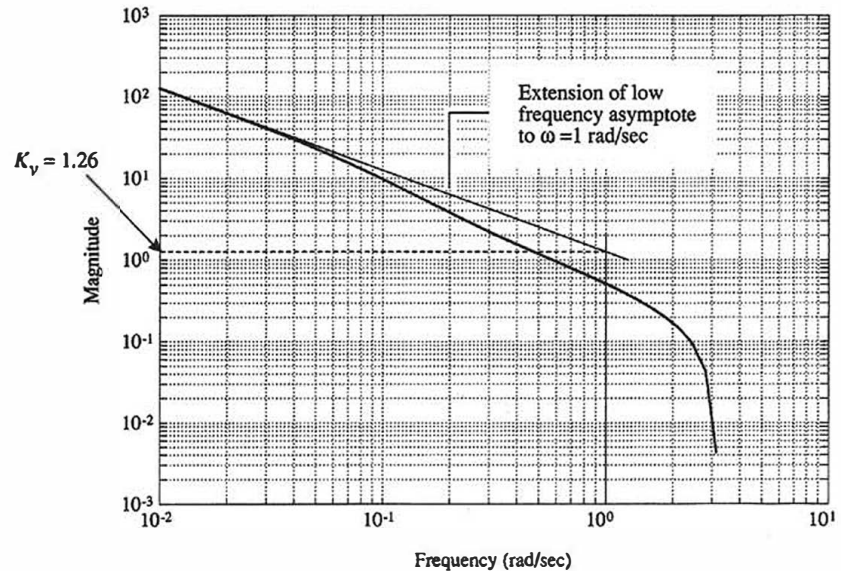
Use the discrete Bode plot to determine the K_v for the antenna system of Example 7.6 with the compensation given by Eq. (7.19).

Solution. The open-loop discrete transfer function is

$$G(z)D(z) = (0.0484) \frac{z + 0.9672}{(z-1)(z-0.9048)} \stackrel{(6)}{=} \frac{z-0.80}{z-0.05},$$

which yields the magnitude versus frequency in Fig. 7.32. Evaluation of the magnitude of the low-frequency asymptote at $\omega = 1$ indicates that $K_v = 1.26$. Also note in the figure that the

Figure 7.32
Determination of K_v
from frequency response



extension of the low-frequency asymptote reaches crossover at $\omega = 1.26$, thus indicating also that $K_v = 1.26$ since this Type 1 system has a low-frequency slope of -1 .

7.4.4 Compensator Design

The amplitude and phase curves can be used to determine the stability margins based on the Nyquist stability criterion for either continuous or discrete systems. In the continuous case with minimum-phase transfer functions, Bode showed that the phase is uniquely determined by an integral of the slope of the magnitude curve on a log-log plot as expressed by Eq. (7.47). If the function is rational, these slopes are readily and adequately approximated by constants! As a consequence the amplitude curve must cross unity gain (zero on a log scale) at a slope of -1 for a reasonable phase margin. The ability to predict stability from the amplitude curve alone in minimum phase systems is an important contributor to the ease with which designers can evaluate changes in compensator parameters in those cases.

Bode's relationship between the amplitude and phase curve is lost for discrete systems because the variable z takes on values around the unit circle in contrast to s traversing the imaginary axis as in continuous systems. Figure 7.16 illustrates the degree to which the relationship is lost and indicates that the error would be small for frequencies lower than 1/20th of the sample frequency. However, it is typically

necessary to determine both magnitude and phase for discrete z -plane systems and not depend on magnitude alone for an accurate assessment of the stability. In carrying out direct digital designs, some intuition from continuous design can be used if the z -plane poles and zeros on the real axis are measured by their distance from $+1$. For example, the equivalent idea in the z -plane for the “breakpoint” in Bode’s hand-plotting rules is that the magnitude will change slope at a frequency when ωT , the *angular* position on the unit circle in radians, has the same value as the fractional distance of the singularity on the real axis to $z = +1$. For example, a pole at $z = 0.9$ will produce a slope change at $\omega T = 0.1$ rad. This equivalence is very accurate for low angular values ($\omega T \leq 0.1$ rad, i.e., sampling at more than 60 times the frequency) and is a reasonable approximation for angular values less than 0.8 rad (i.e., sampling at more than 8 times the frequency). In order to arrive at trial compensations with potential for better *PM*, *GM*, steady-state errors, or crossover frequency, it is useful to understand how a pole or zero placement will affect the magnitude and phase curves. Because of the equivalence of the break-point concept between the continuous and discrete cases, this is accomplished for discrete systems using the ideas from the continuous Bode hand-plotting techniques, keeping in mind that their fidelity degrades as frequency approaches the Nyquist frequency. It is easiest to select compensator break points if the frequency-response magnitude and phase is plotted versus ωT so that the correspondence between those curves and the location of the compensation parameters is retained.

◆ Example 7.15 Design of the Antenna Servo Control

Design a discrete controller for the antenna system with $T = 1$ using the frequency response. The specifications are as before: overshoot less than 16%, settling time less than 10 sec and $K_v \geq 1$.

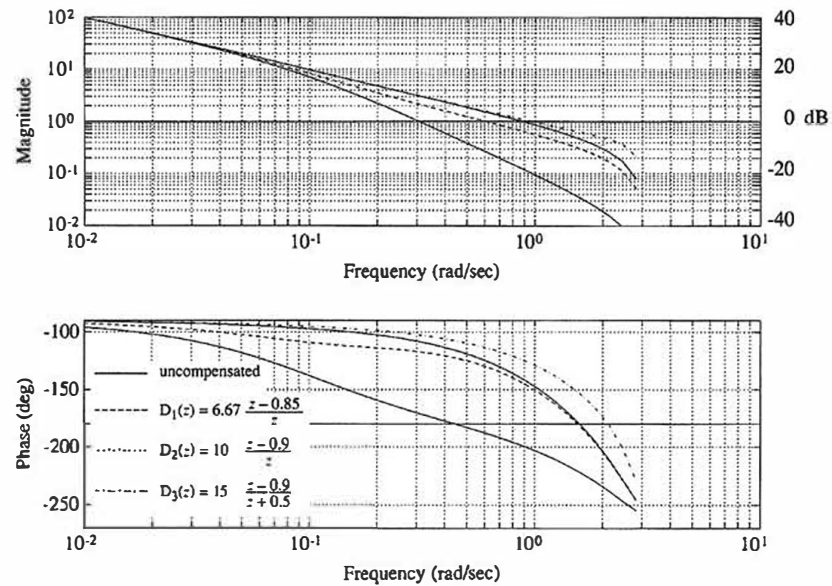
Solution. The system transfer function is

$$G(z) = 0.0484 \frac{z + 0.9672}{(z - 1)(z - 0.9048)} \quad (7.53)$$

The magnitude and phase of the uncompensated system $[G(z)]$ shown in Fig. 7.35 indicate that with a compensator gain of $K = 1$ the system has a *PM* of 8° and a gain crossover frequency (ω_{cg}) of 0.3 rad/sec. The 16% overshoot requirement translates into $\zeta \geq 0.5$, which translates in turn into the requirement that the *PM* be $\geq 50^\circ$ from Fig. 7.24. The specification for settling time translates into the requirement that $\omega_n \geq 0.92$.

Because the gain of $(z - 1)G(z)$ at $z = 1$ is 1, and $T = 1$, the compensated system will also have $K_v = 1$ provided the DC gain of $D(z) = 1$. In terms of the frequency response, this means that the extension of the low-frequency-magnitude asymptote should pass through the value 1 at $\omega = 1$ for the uncompensated case (in Fig. 7.33), and the gain of this low-frequency asymptote should not be decreased with any candidate compensation. To maintain an acceptable K_v , we will evaluate only $D(z)$'s with a DC gain of 1. The uncompensated

Figure 7.33
Frequency responses
with D_1 , D_2 , and D_3 for
Example 7.15



system's PM of 8° indicates poor damping, and the ω_{cg} of 0.3 rad/sec indicates that it will be too slow. Just as for continuous systems, ω_{cg} occurs approximately at the system bandwidth and dominant natural frequency; therefore, we should try to change the design so that it has a ω_{cg} of about 0.9 rad/sec in order to meet the $t_s \leq 10$ sec. Once we find a compensation that meets the guidelines of $PM = 50^\circ$ and $\omega_{cg} = 0.9$ rad/sec, we will need to check whether the settling time and overshoot specifications are actually met, because the design guidelines followed are only approximate.

Figure 7.33 shows several attempts to produce a design. The breakpoint of the first attempt ($D_1(z)$ in Fig. 7.33) was at 0.15 rad/sec⁴ and did not increase the slope of the magnitude curve at a low enough frequency to bring about the desired ω_{cg} . This was remedied in $D_2(z)$, where the breakpoint was lowered to 0.1 rad/sec (zero at $z = 0.9$) causing a ω_{cg} of 0.9 rad/sec, but the resulting PM of 40° was still lower than desired. By moving the compensator pole out to $z = -0.5$ in $D_3(z)$, we had very little effect on the ω_{cg} but achieved an increase in the PM to 50° . Because both goals are met, $D_3(z)$ has a reasonable chance to meet the specifications; in fact, the calculation of a time history of the system response to a step input shows that the t_s is 7 sec, but, alas, the overshoot is 27%. The guidelines were not successful in meeting the specifications because the system is third order with a zero, whereas the rules were derived assuming a second-order system without a zero.

The necessary revisions to our design guidelines are clear; we want more than a 50° PM and do not require a 0.9 rad/sec ω_{cg} . Figure 7.34 shows the system frequency response using $D_3(z)$ along with two revisions of $\bar{D}(z)$ that satisfy our revised goals. $D_4(z)$ has a 60° PM and a 0.6 rad/sec ω_{cg} , and $D_5(z)$ has a 58° PM and a 0.8 rad/sec ω_{cg} . The time history of the system

⁴ The zero at $z = 0.85$ translates into a 0.15 rad/sec breakpoint only because the sample period, T , is 1 sec. For $T = 0.1$ sec, a zero at $z = 0.85$ would translate into a 1.5 rad/sec breakpoint, etc.

Figure 7.34
Bode plots of designs
with controllers D_3 , D_4 ,
and D_5 for Example 7.15

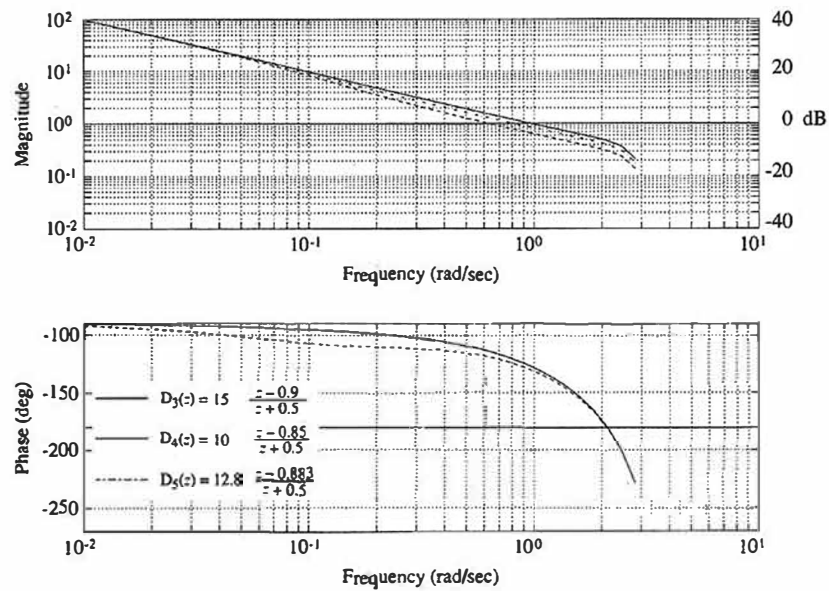
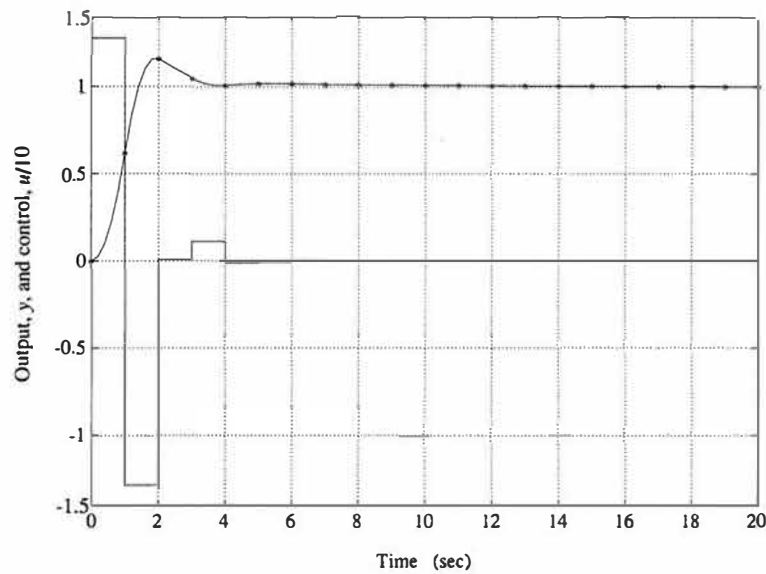
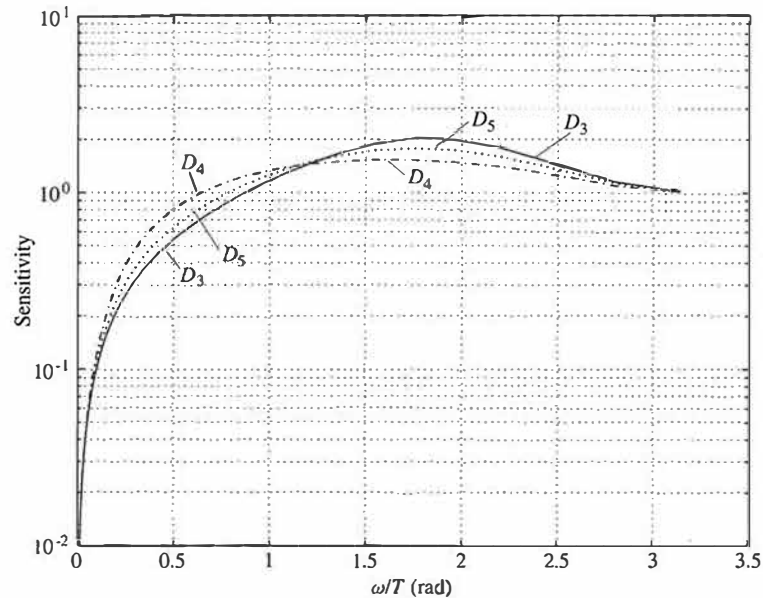


Figure 7.35
Step response of the
system with controller D_5



response to a step using $D_5(z)$ in Fig. 7.35 shows that it exactly meets the requirements for 16% overshoot and $t_s \approx 10$ sec. Furthermore, the design of the system imposed the constraint that $K_v = 1$ and the design is complete.

Figure 7.36
Sensitivity plots of
designs with controllers
 D_3 , D_4 , and D_5 for
Example 7.15



It is interesting to look at the sensitivity functions for these designs, plotted in Fig. 7.36 as log of the sensitivity versus a linear frequency scale to illustrate the balance between positive and negative areas for such plots for stable systems. On this plot, one can see that controller D_3 results in the highest bandwidth and also the highest maximum of the sensitivity or lowest vector gain margin. Controller D_4 has improved robustness (lower maximum of the sensitivity) but also lower bandwidth. Finally, the design given by D_5 splits the difference and meets each specification.

7.5 Direct Design Method of Ragazzini

Much of the style of the transform design techniques we have been discussing in this chapter grew out of the limitations of technology that was available for realization of continuous-time compensators with pneumatic components or electric networks and amplifiers. In particular, many constraints were imposed in order to assure the realization of electric compensator networks $D(s)$ consisting only of resistors and capacitors.⁵ With controllers realized by digital computer, such limitations on realization are, of course, not relevant and one can ignore these particular constraints. An alternative design method that ignores constraints of

⁵ In the book by Truxal (1955), where much of this theory is collected at about the height of its first stage of development, a chapter is devoted to RC network synthesis.

technology has been found to be useful in adaptive controls. Suppose we are given a discrete transfer function $G(z)$ of the plant (plus hold) and a desired transfer function $H(z)$ between reference R and output Y . The structure is assumed to be a unity feedback system and the design is to select the computer transfer function $D(z)$ to realize $H(z)$. The overall transfer function is given by the formula

$$H(z) = \frac{DG}{1 + DG},$$

from which we get the direct design formula

$$D(z) = \frac{1}{G(z)} \frac{H(z)}{1 - H(z)}. \quad (7.54)$$

From Eq. (7.54) we can see that this design calls for a $D(z)$ that will cancel the plant effects and add whatever is necessary to give the desired result. The problem is to discover and implement constraints on $H(z)$ so that we do not ask for the impossible.

First, the design must be causal. From z -transform theory we know that if $D(z)$ is to be causal, then as $z \rightarrow \infty$ its transfer function is bounded and does not have a pole at infinity. Looking at Eq. (7.54), we see that if $G(z)$ were to have a zero at infinity, then $D(z)$ would have a pole there unless we require an $H(z)$ that cancels it. Thus we have the constraint that for $D(z)$ to be causal

$$\begin{aligned} H(z) \text{ must have a zero at infinity of the same order} \\ \text{as the zero of } G(z) \text{ at infinity.} \end{aligned} \quad (7.55)$$

This requirement has an elementary interpretation in the time domain: A zero of order k at infinity in $G(z)$ corresponds to a delay of k samples in the pulse response of the plant. The causality requirement on $H(z)$ is that the closed-loop system must have at least as long a delay as the plant.

Considerations of stability add a second constraint. The roots of the characteristic equation of the closed-loop system are the roots of the equation

$$1 + D(z)G(z) = 0. \quad (7.56)$$

We can express Eq. (7.56) as a polynomial if D and G are rational and we identify $D = c(z)/d(z)$ and $G = b(z)/a(z)$ where a , b , c , and d are polynomials. Then the characteristic polynomial is

$$ad + bc = 0. \quad (7.57)$$

Now suppose there is a common factor in DG , as would result if $D(z)$ were called upon to cancel a pole or zero of $G(z)$. Let this factor be $z - \alpha$ and suppose it is a pole of $G(z)$, so we can write $a(z) = (z - \alpha)\bar{a}(z)$, and to cancel it we have $c(z) = (z - \alpha)\bar{c}(z)$. Then Eq. (7.57) becomes

$$\begin{aligned} (z - \alpha)\bar{a}(z)d(z) + b(z)(z - \alpha)\bar{c}(z) &= 0, \\ (z - \alpha)[\bar{a}d + b\bar{c}] &= 0. \end{aligned} \quad (7.58)$$

In other words—perhaps it was obvious from the start—a common factor *remains a factor of the characteristic polynomial*. If this factor is outside the unit circle, the system is unstable! How do we avoid such cancellation? Considering again Eq. (7.54), we see that if $D(z)$ is not to cancel a pole of $G(z)$, then that factor of $a(z)$ must also be a factor of $1 - H(z)$. Likewise, if $D(z)$ is not to cancel a zero of $G(z)$, such zeros must be factors of $H(z)$. Thus we write the constraints⁶

$$1 - H(z) \text{ must contain as zeros all the poles of } G(z) \text{ that are outside the unit circle,} \quad (7.59)$$

$$H(z) \text{ must contain as zeros all the zeros of } G(z) \text{ that are outside the unit circle.} \quad (7.60)$$

Consider finally the constraint of steady-state accuracy. Because $H(z)$ is the overall transfer function, the error transform is given by

$$E(z) = R(z)(1 - H(z)). \quad (7.61)$$

Thus if the system is to be Type 1 with velocity constant K_v , we must have zero steady-state error to a step and $1/K_v$ error to a unit ramp. The first requirement is

$$e(\infty) = \lim_{z \rightarrow 1} (z - 1) \frac{1}{z - 1} [1 - H(z)] = 0, \quad (7.62)$$

which implies

$$H(1) = 1. \quad (7.63)$$

The velocity constant requirement is that

$$e(\infty) = \lim_{z \rightarrow 1} (z - 1) \frac{Tz}{(z - 1)^2} [1 - H(z)] = \frac{1}{K_v}. \quad (7.64)$$

From Eq. (7.63) we know that $1 - H(z)$ is zero at $z = 1$, so that to evaluate the limit in Eq. (7.64), it is necessary to use L'Hôpital's rule with the result

$$-T \left. \frac{dH}{dz} \right|_{z=1} = \frac{1}{K_v}. \quad (7.65)$$

◆ Example 7.16 Design by the Direct Method

Consider again the plant described by the transfer function of Eq. (7.53) and suppose we ask for a digital design that has the characteristic equation that is the discrete equivalent of the continuous characteristic equation

$$s^2 + s + 1 = 0,$$

with a sampling period $T = 1$ sec.

⁶ Roots on the unit circle are also unstable by some definitions, and good practice indicates that we should not cancel singularities outside the radius of desired settling time.

Solution. The discrete characteristic equation according to the specifications is

$$z^2 - 0.7859z + 0.36788 = 0. \quad (7.66)$$

Let us therefore ask for a design that is stable, has $K_v = 1$, and has poles at the roots of Eq. (7.66) plus, if necessary, additional poles at $z = 0$, where the transient is as short as possible. The form of $H(z)$ is thus

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + \dots}{1 - 0.7859z^{-1} + 0.3679z^{-2}}. \quad (7.67)$$

The causality design constraint, using Eq. (7.55) requires that

$$H(z)|_{z=\infty} = 0$$

or

$$b_0 = 0. \quad (7.68)$$

Equations (7.59) and (7.60) add no constraints because $G(z)$ has all poles and zeros inside the unit circle except for the single zero at ∞ , which is taken care of by Eq. (7.68). The steady-state error requirement leads to

$$\begin{aligned} H(1) &= 1 \\ &= \frac{b_1 + b_2 + b_3 + \dots}{1 - 0.7859 + 0.3679} = 1. \end{aligned} \quad (7.69)$$

Therefore

$$b_1 + b_2 + b_3 + \dots = 0.5820,$$

and

$$-T \left. \frac{dH}{dz} \right|_{z=1} = \frac{1}{K_v}.$$

Because in this case both T and K_v are 1, we use Eq. (7.69) and the derivative with respect to z^{-1} to obtain

$$\begin{aligned} 1 &= \frac{1}{K_v} = \left. \frac{dH}{dz^{-1}} \right|_{z=1} \\ &= \frac{(0.5820)[b_1 + 2b_2 + 3b_3 + \dots] - [0.5820][-0.7859 + 0.3679(2)]}{(0.5820)(0.5820)} \end{aligned}$$

or

$$\frac{b_1 + 2b_2 + 3b_3 + \dots - [-0.05014]}{0.5820} = 1. \quad (7.70)$$

Because we have only two equations to satisfy, we need only two unknowns and we can truncate $H(z)$ at b_2 . The resulting equations are

$$b_1 + b_2 = 0.5820,$$

$$b_1 + 2b_2 = 0.5318,$$

which have the solution

$$b_1 = 0.6321, \quad (7.71)$$

$$b_2 = -0.05014. \quad (7.72)$$

Thus the final design gives an overall transfer function

$$H(z) = \frac{0.6321z - 0.05014}{z^2 - 0.7859z + 0.3679}. \quad (7.73)$$

We shall also need

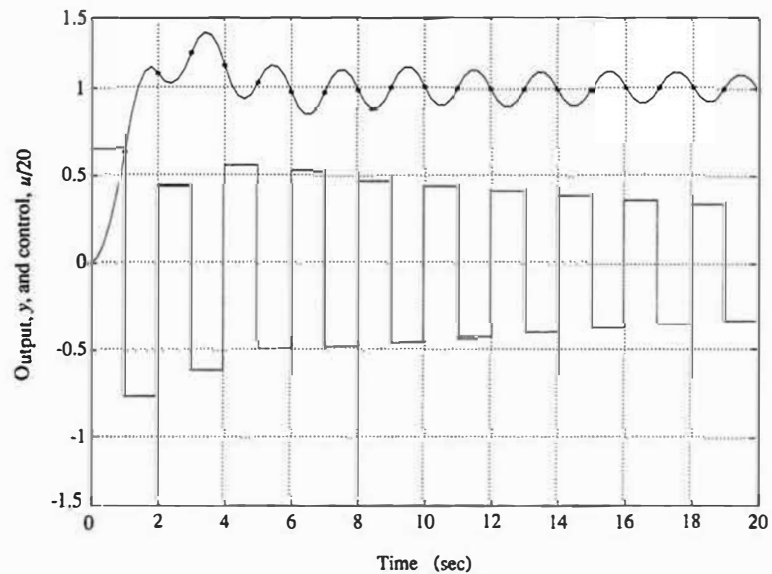
$$1 - H(z) = \frac{(z - 1)(z - 0.4180)}{z^2 - 0.7859z + 0.3679}. \quad (7.74)$$

We know that $H(1) = 1$ so that $1 - H(z)$ must have a zero at $z = 1$. Now, turning to the basic design formula, Eq. (7.54), we compute

$$\begin{aligned} D(z) &= \frac{(z - 1)(z - 0.9048)(0.6321)}{(0.04837)(z + 0.9672)} \frac{(z - 0.07932)}{(z - 1)(z - 0.4180)} \\ &= 13.07 \frac{(z - 0.9048)(z - 0.07932)}{(z + 0.9672)(z - 0.4180)}. \end{aligned}$$

A plot of the step response of the resulting design for this example is provided in Fig. 7.37 and verifies that the response samples behave as specified by $H(z)$. However, as can be seen also from the figure, large oscillations occur in the

Figure 7.37
Step response of
antenna system from
direct design



control that cause a large ripple in the system response between samples. How can this be for a system response transfer function,

$$\frac{Y(z)}{R(z)} = H(z) = \frac{DG}{1 + DG},$$

that is designed to have only two well-damped roots? The answer lies in the fact that the control response is determined from

$$\frac{U(z)}{R(z)} = \frac{D}{1 + DG} = \frac{H(z)}{G(z)},$$

which for this example is

$$\frac{U(z)}{R(z)} = 13.06 \frac{z - 0.0793}{z^2 - 0.7859z + 0.3679} \frac{(z - 1)(z - 0.9048)}{z + 0.9672}.$$

The pole at $z = -0.9672$, very near to the unit circle, is the source of the oscillation in the control response. The poor transient due to the pole did not show up in the output response because it was exactly canceled by a zero in the plant transfer function. The large control oscillation in turn causes the ripple in the output response. This pole was brought about because we allowed the controller to have a pole to cancel a plant zero at this position. The poor response that resulted could have been avoided if this nearly unstable zero had been included in the stability constraint list. In that case we would introduce another term in $H(z)$, $b_3 z^{-3}$, and require that $H(z)$ be zero at $z = -0.9672$, so this zero of $G(z)$ is not canceled by $D(z)$. The result will be a simpler $D(z)$ with a slightly more complicated $H(z)$.

7.6 Summary

In this chapter we have reviewed the philosophy and specifications of the design of control systems by transform techniques and discussed three such methods.

- Discrete controllers can be designed by emulation, root locus, or frequency response methods.
- Successful design by emulation typically requires a sampling frequency at least 30 times the expected closed-loop bandwidth.
- Expressions for steady-state error constants for discrete systems have been given in Eq. (7.12) and Eq. (7.14) in terms of open-loop transfer functions and in Eq. (7.18) in terms of closed-loop poles and zeros.
- Root locus rules for discrete system characteristic equations are shown to be the same as the rules for continuous system characteristic equations.
- Step response characteristics such as rise time and overshoot can be correlated with regions of acceptable pole locations in the z -plane as sketched in Fig. 7.10.

- Asymptotes as used in continuous system frequency response plots *do not apply* for discrete frequency response.
- Nyquist's stability criterion and gain and phase margins were developed for discrete systems.
- The sensitivity function was shown to be useful to develop specifications on performance robustness as expressed in Eq. (7.34).
- Stability robustness in terms of the overall transfer function, the complementary sensitivity function, is expressed in Eq. (7.47).
- Limitations on the frequency response of closed-loop discrete designs are made more severe by poles and zeros outside the unit circle as expressed by Eq. (7.50) and Eq. (7.51).
- Lead and lag compensation can be used to improve the steady-state and transient response of discrete systems.
- The direct design method of Ragazzini can be used to realize a closed-loop transfer function limited only by causality and stability constraints.

7.7 Problems

- 7.1 Use the $z = e^{sT}$ mapping function and prove that the curve of constant ζ in s is a logarithmic spiral in z .
- 7.2 A servomechanism system is expected to have a rise-time of no more than 10 milliseconds and an overshoot of no more than 5%.
- Plot in the s -plane the corresponding region of acceptable closed-loop pole locations.
 - What is the estimated Bode gain crossover frequency (rad/sec)?
 - What is the estimated phase margin in degrees?
 - What is the sample period, T , if the estimated phase shift due to the sample and hold is to be no more than 10° at the gain crossover?
 - What is the sample period T if there are to be 8 samples per rise time?
- 7.3 *Root locus review:* The following root loci illustrate important features of the root locus technique. All are capable of being sketched by hand, and it is recommended that they be done that way in order to develop skills in verifying a computer's output. Once sketched roughly by hand, it is useful to fill in the details with a computer.
- The locus for

$$1 + K \frac{s + 1}{s^2(s + p_1)} = 0$$

is typical of the behavior near $s = 0$ of a double integrator with lead compensation or a single integration with a lag network and one additional real pole. Sketch the locus versus K for values of p_1 of 5, 9, and 20. Pay close attention to the real axis break-in and break-away points.

- (b) The locus for

$$1 + K \frac{1}{s(s+1)((s+a)^2+4)} = 0$$

is a typical locus that includes complex poles and shows the value of departure angles. Plot the locus for $a = 0, 1$, and 2 . Be sure to note the departure angles from the complex poles in each case.

- (c) The locus for

$$1 + K \frac{(s+1)^2 + \omega_o^2}{s(s^2+4)} = 0$$

illustrates the use of complex zeros to compensate for the presence of complex poles due to structural flexibility. Be sure to estimate the angles of departure and arrival. Sketch the loci for $\omega_o = 1$ and $\omega_o = 3$. Which case is unconditionally stable (stable for all positive K less than the design value)?

- (d) For

$$1 + K \frac{s}{(s-p_1)(s-p_2)} = 0$$

show that the locus is a circle of radius $\sqrt{p_1 p_2}$ centered at the origin (location of the zero). Can this result be translated to the case of two poles and a zero on the negative real axis?

- 7.4 The basic transfer function of a satellite attitude control is $G(s) = \frac{1}{s^2}$.

- (a) Design a continuous lead network compensation so as to give closed-loop poles corresponding to $\zeta = 0.5$ and natural frequency $\omega_n = 1.0$. The ratio of pole to zero of the lead is to be no more than 10.
 - i. Plot the step response of the design and note the rise time and the percent overshoot.
 - ii. What is the system type and corresponding error constant?
- (b) Select a sampling period to give 10 samples in a rise time and compute the discrete equivalent to the lead using the Tustin bilinear transformation. Plot the step response of the discrete system and compare the rise time and overshoot to those of the continuous design.
- (c) Select a sampling period that will give 5 samples per rise time, compute the discrete equivalent using Tustin's method, and compare rise time and overshoot of this design with the continuous case.

- 7.5 Repeat the design for the satellite attitude control of Problem 4, including method of choosing sampling periods but using the matched pole-zero method to obtain the discrete compensations.

- 7.6 Repeat the design of the satellite attitude control of Problem 4 including method of choice of sampling period but using the triangle hold equivalent (noncausal first-order hold) to design the discrete compensations.

- 7.7 Repeat the design for the satellite attitude control of Problem 4 but augment the plant with a Pade approximation to the delay of $T/2$ which is to say, multiply the plant transfer function by

$$P(s) = \frac{1 - \frac{sT}{4}}{1 + \frac{sT}{4}}$$

before doing the continuous design. Once the design of the lead compensation is completed, continue with the discrete equivalents as in Problem 4, including the method of choosing sampling periods. Use the matched pole-zero method to obtain the discrete compensations. Compare the design with the continuous case.

- 7.8 Design a discrete compensation for the antenna control system as specified in Example 7.2 with a sample period of $T = 0.1$ using a matched pole-zero equivalent for the discrete compensation. Plot the step response and compare rise time and overshoot with those of the continuous design.
- 7.9 Design the antenna control system as specified in Example 7.2 with a sample period of $T = 0.5$ sec.
- Use the zero-pole mapping equivalent emulation method.
 - Augment the plant model with an approximation of the sample-hold delay consisting of

$$P(s) = \frac{2/T}{s + 2/T},$$

then redesign $D(s)$ and find the discrete equivalent with the matched pole-zero equivalent emulation method. Plot the step response and compare with the continuous design done on the unaugmented plant.

- Compare the degradation of the equivalent damping ratio ζ due to sampling for both design methods.
- 7.10 For the satellite with transfer function $1/s^2$, design a lead compensation to give closed-loop poles with damping $\zeta = 0.5$ and natural frequency $\omega_n = 1.0$. The pole-to-zero ratio of the compensation should not be more than 10. Plot the step response of the closed loop and note the rise time and overshoot.
- Let sampling period be $Ts = 0.5$ sec and compute the discrete model of the plant with a sample and zero-order hold. Using this model, design a discrete lead compensation with pole at $z = -0.5$ and a zero so as to give closed loop poles at the mapped place from the continuous poles, $\zeta = 0.5$ and $\omega_n = 1.0$. What is the ratio of ω_s/ω_n for this problem? How many samples do you expect to find per rise time? Plot the step response and compare result with expectation. Compare the discrete design with the continuous design.
 - Repeat the discrete design with sampling period $Ts = 1.2$ sec, plot the step response, and compare rise time and overshoot with the continuous case.
- 7.11 Sketch the region in the z -plane of discrete pole locations corresponding to $\zeta \geq 0.5$ and
- $\omega_n \leq \omega_s/30$.
 - $\omega_n \leq \omega_s/10$.
 - $\omega_n \leq \omega_s/5$.

7.12 The plant transfer function

$$G(s) = \frac{1}{(s + 0.1)(s + 3)}$$

is to be controlled with a digital controller using a sample period of $T = 0.1$ sec.

- Design compensation using the z -plane root locus that will respond to a step with a rise time of ≤ 1 sec and an overshoot $\leq 5\%$. Plot the step response and verify that the response meets the specifications.
- What is the system type and corresponding error constant? What can be done to reduce the steady-state error to a step input?
- Design a discrete lag compensation that will cut the steady-state error in half. Plot the step response and compare the complete response to the transient and steady-state error specifications.

7.13 It is possible to suspend a steel ball bearing by means of an electromagnet whose current is controlled by the position of the mass [Woodson and Melcher (1968)]. A schematic of a possible setup is shown in Fig. 7.38. The equations of motion are

$$m \ddot{X} = -mg + f(X, I),$$

where the force on the ball due to the electromagnet is given by $f(X, I)$. It is found that the magnet force balances the gravity force when the magnet current is I_0 and the ball at X_0 . If we write $I = I_0 + i$ and $X = X_0 + x$ and expand f about $X = X_0$ and $I = I_0$, and then neglect higher-order terms, we obtain a linear approximation

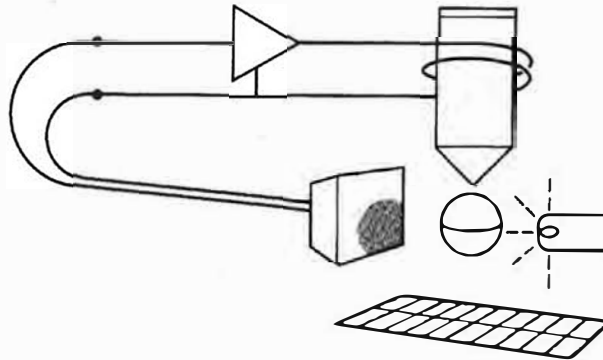
$$m \ddot{x} = k_1 x + k_2 i.$$

Values measured for a particular device in the Stanford Controls Laboratory are $m = 0.02$ kg, $k_1 = 20$ N/m, $k_2 = 0.4$ N/A.

- Compute the transfer function from i to x and draw the (continuous) root locus for proportional feedback $i = -Kx$.
- Let the sample period be $T = 0.02$ sec and compute the plant discrete transfer function when used with a sample and zero-order hold.
- Design a digital control for the magnetic levitation to meet the specifications $t_r \leq 0.1$ sec, $t_s \leq 0.4$ sec, and overshoot $\leq 20\%$.

Figure 7.38

A steel ball balanced by an electromagnet



- (d) Plot a root locus of your design versus m and discuss the possibility of balancing balls of various masses.
- (e) Plot a step response of your design to an initial disturbance displacement on the ball and show both x and the control current i . If the sensor can measure x over a range of only $\pm \frac{1}{2}$ cm, and if the amplifier can provide a maximum current of 1 A, what is the *maximum* initial displacement, $x(0)_{\max}$ that will keep the variables within these limits, using $m = 0.02$ kg?

7.14 A discrete transfer function for approximate derivative control is

$$D(z) = K_p T \frac{z-1}{D T z},$$

where the pole at $z = 0$ adds some destabilizing phase lag. It therefore seems that it would be advantageous to remove it and to use derivative control of the form

$$D(z) = K_p T_D \frac{(z-1)}{T}.$$

Can this be done? Support your answer with the difference equation that would be required and discuss the requirements to implement it.

7.15 For the automotive cruise-control system shown in Fig. 7.39, the sample period is $T = 0.5$ sec.

- (a) Design a PD controller to achieve a t_r of 5 sec with no overshoot.
- (b) Determine the speed error on a 3% grade (i.e., $G_r = 3$ in Fig. 7.39).
- (c) Design a PID controller to meet the same specifications as in part (a) and that has zero steady-state error on constant grades. What is the velocity constant of your design?

7.16 For the disk drive read/write head assembly described in Fig. 7.40, you are to design a compensation that will result in a closed-loop settling time $t_s = 20$ msec and with overshoot to a step input $M_p \leq 20\%$.

- (a) Assume no sampling and use a continuous compensation. Plot the step response and verify that your design meets the specifications.
- (b) Assume a sampling period $T = 1$ msec and use matched pole-zero emulation. If you wish, you can include a Pade approximation to the delay and do a redesign of the continuous compensation before computing the discrete equivalent. Plot the step response and compare it with the continuous design's response.

Figure 7.39
An automotive
cruise-control system

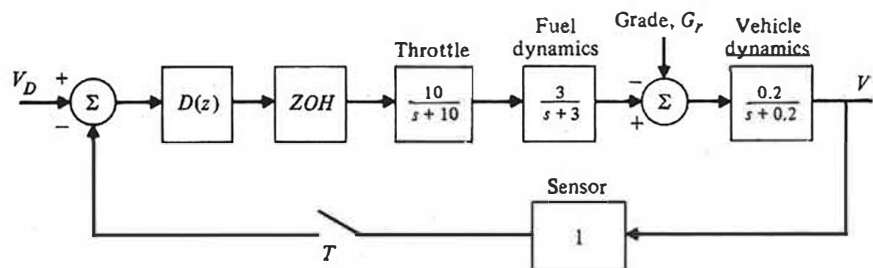
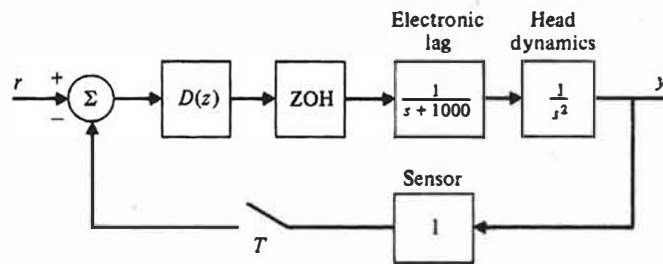


Figure 7.40
A disk drive read/write head assembly



- (c) Do a z-plane design for the same specifications and plot its step response. Compare these three designs with respect to meeting the transient performance.
- 7.17 The tethered satellite system shown in Fig. 7.41 has a moveable tether attachment point so that torques can be produced for attitude control. The block diagram of the system is shown in Fig. 7.42. Note that the integrator in the actuator block indicates that a constant-voltage command to the servomotor will produce a constant velocity of the attachment point.
- Is it possible to stabilize this system with θ feedback to a PID controller? Support your answer with a root locus argument.
 - Suppose it is possible to measure δ , and $\dot{\theta}$ as well as θ . Select the variable(s) on which you would like to place a sensor(s) to augment the θ feedback.
 - Design compensation for the system using the sensor(s) that you selected in part (b) so that it has a 2-sec rise time and equivalent closed loop damping of $\zeta = 0.5$.

Figure 7.41
A tethered satellite system

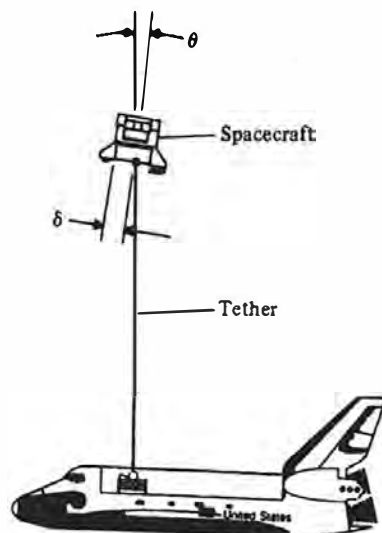
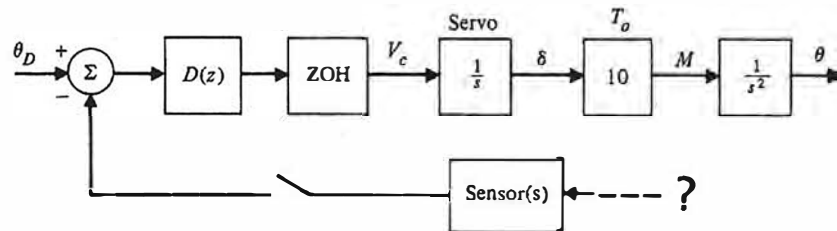


Figure 7.42

Block diagram for the tethered satellite system

 δ = displacement of tether attachment point T_o = tether tension M = moment on satellite θ = satellite attitude**Figure 7.43**

An excavator with an automatic control system



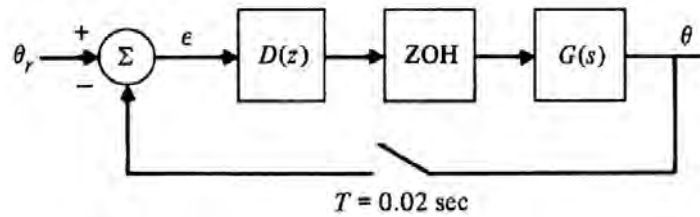
- 7.18** The excavator shown in Fig. 7.43 has a sensor measuring the angle of the stick as part of a control system to control automatically the motion of the bucket through the earth. The sensed stick angle is to be used to determine the control signal to the hydraulic actuator moving the stick. The schematic diagram for this control system is shown in Fig 7.44, where $G(s)$ is the system transfer function given by

$$G(s) = \frac{1000}{s(s + 10)(s^2 + 1.2s + 144)}$$

The compensation is implemented in a control computer sampling at $f_s = 50$ Hz and is of the form $D(z) = K(1 + K_1(1 - z^{-1}) + K_2(1 - 2z^{-1} + z^{-2}))$. The oscillatory roots in $G(s)$ arise from the compressibility of the hydraulic fluid (with some entrained air) and is often referred to as the *oil-mass resonance*.

Figure 7.44

Schematic diagram for the control system of the excavator



- (a) Show that the steady-state error, $\epsilon (= \theta_r - \theta)$, is

$$\epsilon(\infty) = \frac{1.44}{K},$$

when θ_r is a unit ramp.

- (b) Determine the highest K possible (i.e., at the stability boundary) for proportional control ($K_1 = K_2 = 0$).
- (c) Determine the highest K possible (i.e., at the stability boundary) for PD ($K_2 = 0$) control.
- (d) Determine the highest K possible (i.e., at the stability boundary) for PD plus acceleration ($K_2 \neq 0$) control.⁷

7.19 For the excavator described in Problem 7.18 with transfer function

$$G(s) = \frac{1000}{s(s+10)(s^2+1.2s+144)},$$

plot the Bode plot and measure the gain and phase margins with just unity feedback.

- (a) Design a compensation that will give a phase margin of 50° , a gain margin as measured at the resonance peak of at least 2 and a crossover of at least $\omega_{cg} \geq 1.0$. Plot the step response of the resulting design and note the rise time and the overshoot.
- (b) With sample frequency $f_s = 50 \text{ Hz}$, design a discrete controller to meet the same specifications given for the continuous case. Plot the Bode plot of the design and verify that the specifications are met.

7.20 A simple model of a satellite attitude control has the transfer function $1/s^2$.

- (a) Plot the Bode plot for this system and design a lead compensation to have a phase margin of 50° and a crossover ω_{cg} of 1.0 rad/sec. Plot the step response and note the rise time and overshoot.
- (b) With sample period of $T = 0.2$ sec, design a discrete compensation to give 50° phase margin and crossover $\omega_{cg} = 1.0$. Plot the step response and compare rise time and overshoot with the continuous design.
- (c) With sample period $T = 0.5$ sec, design a discrete compensation to give 50° phase margin and crossover $\omega_{cg} = 1.0$. Plot the step response and compare rise time and overshoot with the continuous design.

7.21 For a system given by

$$G(s) = \frac{a}{s(s+a)},$$

⁷ For further reading on damping the oil-mass resonance of hydraulic systems, see Viersma (1980).

determine the conditions under which the K_v of the continuous system is approximately equal to the K_v of the system preceded by a ZOH and represented by its discrete transfer function.

7.22 Design a digital controller for

$$G(s) = \frac{1}{s(s + 0.4)}$$

preceded by a ZOH so that the response has a rise time of approximately 0.5 sec, overshoot $< 25\%$, and zero steady-state error to a step command. [Hint: Cancel the plant pole at $s = 0.4$ with a compensator zero; a second-order closed-loop system will result, making the transient response comparison between experiment and theory much easier.]

- (a) Determine a $D(z)$ using emulation with the matched pole-zero mapping technique. Do two designs, one for $T = 100$ msec and one for $T = 250$ msec.
- (b) Repeat part (a) using the z -plane root locus method for the two sample periods.
- (c) Simulate the closed-loop system response to a unit step with the $D(z)$'s obtained in parts (a) and (b). Use the discrete equivalent of the plant in your calculations. Compare the four digitally controlled responses with the original specifications. Explain any differences that you find.

7.23 A generic mechanical control problem is that of two masses coupled by a lightly damped flexible structure. With amplitude and time scaling, the model can be reduced to

$$G(s) = K \frac{Bs + 1}{s^2(s^2 + Bs + 1)}$$

For the parameters $K = 0.05$ and $B = 0.1$, plot the Bode plot and indicate the gain and phase margins. For performance tracking, the open loop gain for frequencies below $\omega = 0.005$ must be above 50 and the gain for frequencies above $\omega = 1$ must be below 0.5. The phase margin must be at least 35° .

- (a) Design a continuous compensation to meet the specifications. If possible, keep the phase above -180° so the system will not be conditionally stable.
- (b) Design a discrete compensation for the system with the lowest sampling frequency possible. A prize will be given to the student who obtains the lowest sampling frequency with a design that meets the specifications.
- (c) Plot the sensitivity of the digital design and compute the vector gain margin.

• 8 •

Design Using State-Space Methods

A Perspective on State-Space Design Methods

In Chapter 7, we discussed how to design digital controllers using transform techniques, methods now commonly designated as “classical design.” The goal of this chapter is to solve the identical problem using the state-space formulation. The difference in the two approaches is entirely in the design method; the end result, a set of difference equations providing control, is identical.

Advantages of the state-space formulation are especially apparent when designing controllers for multi-input, multi-output (MIMO) systems, that is, those with more than one control input and/or sensed output. However, state-space methods are also an aid in the design of controllers for single-input, single-output (SISO) systems because of the widespread use of computer-aided control system design (CACSD) tools, which often rely heavily on this system representation. Chapters 4 and 5 have already demonstrated the advantages of the state-space formulation in using CACSD packages for the computation of discrete equivalents. State-space methods also offer advantages in the structure for command inputs and disturbance estimation. In this chapter, we will limit our state-space design efforts to SISO controllers, similar to those found in Chapter 7 with classical methods. Techniques for MIMO design are discussed in Chapter 9.

In Chapter 7, two basic methods were described: *emulation* and *direct digital design*. The same two methods apply to the state-space formulation as well. Using emulation, one would design a continuous controller using state-space methods, then transform the controller to a discrete form by using one of the discrete equivalents from Chapter 6. The discussion of the method and its accuracy in Chapter 7 applies equally well here. Furthermore, the development in Chapter 6 used both classical and state-space system descriptions in the computation of the equivalents. Therefore, no further discussion of emulation is required, and we will concentrate solely on the direct digital design method.

Chapter Overview

As for the continuous case reviewed in Section 2.6, design using state-space involves separate design of the control assuming all state elements are available, then design of an estimator to reconstruct the state given a partial measurement of it. Section 8.1 covers the discrete control design while Section 8.2 covers the discrete estimator design. Section 8.3 puts it together into what is called the **regulator** and Section 8.4 discusses the relative merits of the various ways of introducing the reference input command. Section 8.5 presents how a designer builds an integral control feature or disturbance estimation, and how they accomplish similar goals. Section 8.6 discusses the limitations imposed by delays in the system and how to minimize their effect. The chapter concludes in Section 8.7 with a discussion of observability and controllability, the required conditions for the design to be possible.

8.1 Control Law Design

In Chapter 2, we saw that the state-space description of a continuous system is given by Eq. (2.1)

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}u, \quad (8.1)$$

and Eq. (2.2)

$$y = \mathbf{H}\mathbf{x} + Ju. \quad (8.2)$$

We assume the control is applied from the computer through a ZOH as shown in Fig. 1.1. Therefore, Eqs. (8.1, and (8.2) have an exact discrete representation as given by Eq. (4.59)

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}u(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k) + Ju(k), \end{aligned} \quad (8.3)$$

where

$$\begin{aligned} \mathbf{\Phi} &= e^{\mathbf{F}T}, \\ \mathbf{\Gamma} &= \int_0^T e^{\mathbf{F}\eta} d\eta \mathbf{G}. \end{aligned} \quad (8.4)$$

We can easily transform between the classical transfer function of a continuous system, $G(s)$ (represented by `sysTF` in MATLAB), to the state-space continuous description by the MATLAB script¹

$$\text{sysC} = \text{ss}(\text{sysTF})$$

¹ In MATLAB, $G(s)$ would be represented by `[num,den]` and the `tf2ss` function would be invoked. See Appendix F.

where sysC contains $\mathbf{F}, \mathbf{G}, \mathbf{H}, J$. Provided the continuous system is preceded by a ZOH, we can transform to the discrete model, $\Phi, \Gamma, \mathbf{H}, J$, (or sysD) with a sample period T , using the MATLAB script

$$\text{sysD} = \text{c2d}(\text{sysC}, T).$$

estimator

One of the attractive features of state-space design methods is that the procedure consists of two independent steps. The first step *assumes* that we have all the state elements at our disposal for feedback purposes. In general, of course, this would not be a good assumption; a practical engineer would not, as a rule, find it necessary to purchase such a large number of sensors, especially because he or she knows that they would not be needed using classical design methods. The assumption that all states are available merely allows us to proceed with the first design step, namely, the control law. The remaining step is to design an “estimator” (or “observer”²), which estimates the entire state vector, given measurements of the portion of the state provided by Eq. (8.2). The final control algorithm will consist of a combination of the control law and the estimator with the control-law calculations based on the estimated states rather than on the actual states. In Section 8.3 we show that this substitution is reasonable and that the combined control law and estimator can give closed-loop dynamic characteristics that are unchanged from those assumed in designing the control law and estimator separately. The dynamic system we obtain from the combined control law and estimator is the same that has been previously referred to as compensation.

As for the continuous case, the control law is simply the feedback of a linear combination of all the state elements, that is

$$u = -\mathbf{K}\mathbf{x} = -\begin{bmatrix} K_1 & K_2 & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix}. \quad (8.5)$$

Note that this structure does not allow for a reference input to the system. The topology that we used all through Chapter 7 (Fig. 7.5) always included a reference input, r . The control law, Eq. (8.5), assumes that $r = 0$ and is, therefore, usually referred to as a **regulator**. Section 8.4 will discuss how one introduces reference inputs.

Substituting Eq. (8.5) in Eq. (8.3), we have

$$\mathbf{x}(k+1) = \Phi\mathbf{x}(k) - \Gamma\mathbf{K}\mathbf{x}(k). \quad (8.6)$$

Therefore the z -transform of Eq. (8.6) is

$$(z\mathbf{I} - \Phi + \Gamma\mathbf{K})\mathbf{X}(z) = 0,$$

² The literature [Luenberger (1960)] commonly refers to these devices as “observers”; however, we feel that the term “estimator” is much more descriptive of their function because “observe” implies a direct measurement. In this book the term “estimator” is used but the reader can think of the terms interchangeably.

and the characteristic equation of the system with the hypothetical control law is

$$|zI - \Phi + \Gamma K| = 0. \quad (8.7)$$

8.1.1 Pole Placement

The approach we wish to take at this point is pole placement; that is, having picked a control law with enough parameters to influence all the closed-loop poles, we will arbitrarily select the desired pole locations of the closed-loop system and see if the approach will work. Although this approach can often lead to trouble in the design of complex systems (see the discussion in Sections 2.6 and 8.3), we use it here to illustrate the power of full state feedback. In Chapter 9, we will build on this idea to arrive at a more practical design methodology.

The control-law design, then, consists of finding the elements of $\mathbf{K}$ so that the roots of Eq. (8.7), that is, the poles of the closed-loop system, are in the desired locations. Unlike classical design, where we iterated on parameters in the compensator (hoping) to find acceptable closed-loop root locations, the full state feedback, pole-placement approach guarantees success and allows us to arbitrarily pick any pole location, providing that n poles are specified for an n th-order system.

Given desired pole locations,³ say

$$z_i = \beta_1, \beta_2, \beta_3, \dots, \beta_n,$$

the desired control-characteristic equation is

$$\alpha_c(z) = (z - \beta_1)(z - \beta_2) \cdots (z - \beta_n) = 0. \quad (8.8)$$

Equations (8.7) and (8.8) are both the characteristic equation of the controlled system; therefore, they must be identical, term by term. Thus we see that the required elements of $\mathbf{K}$ are obtained by matching the coefficients of each power of z in Eq. (8.7) and Eq. (8.8), and there will be n equations for an n th-order system.

◆ Example 8.1 Pole Placement for Satellite Attitude Control

Design a control law for the satellite attitude-control system described by Eq. (4.47). Pick the z -plane roots of the closed-loop characteristic equation so that the equivalent s -plane roots have a damping ratio of $\zeta = 0.5$ and real part of $s = -1.8$ rad/sec (i.e., $s = -1.8 \pm j3.12$ rad/sec). Use a sample period of $T = 0.1$ sec.

³ Discussion of how one selects pole locations was reviewed in Section 2.6 for the continuous case and will occur through the following examples and in Section 8.3 for the discrete case. The results of the specification discussion in Chapter 7 can also be used to specify poles.

Solution. Example 4.11 showed that the discrete model for this system is

$$\Phi = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \Gamma = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}.$$

Using $z = e^{sT}$ with a sample period of $T = 0.1$ sec, we find that $s = -1.8 \pm j3.12$ rad/sec translates to $z = 0.8 \pm j0.25$, as shown in Fig. 8.1. The desired characteristic equation is then

$$z^2 - 1.6z + 0.70 = 0, \quad (8.9)$$

and the evaluation of Eq. (8.7) for any control law $\mathbf{K}$ leads to

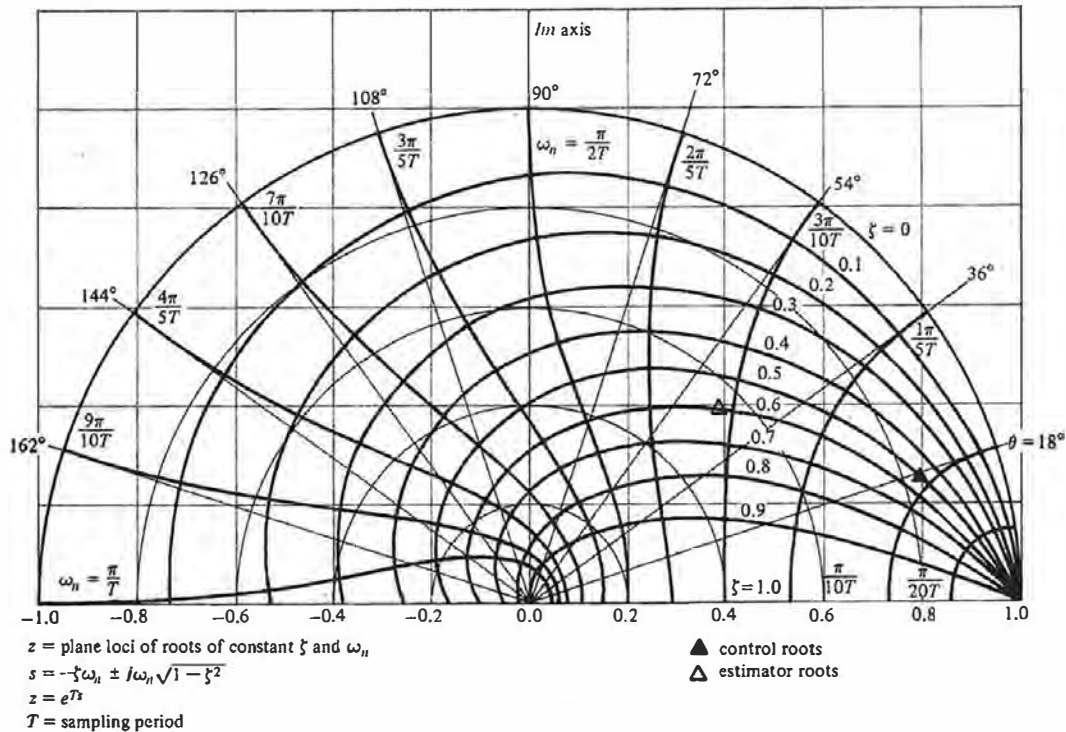
$$\left| z \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} [K_1 \ K_2] \right| = 0$$

or

$$z^2 + (TK_2 + (T^2/2)K_1 - 2)z + (T^2/2)K_1 - TK_2 + 1 = 0. \quad (8.10)$$

Figure 8.1

Desired root locations for satellite attitude-control system of Examples 8.1 and 8.4



Equating coefficients in Eqs. (8.9) and (8.10) with like powers of z , we obtain two simultaneous equations in the two unknown elements of $\mathbf{K}$

$$\begin{aligned}TK_2 + (T^2/2)K_1 - 2 &= -1.6, \\(T^2/2)K_1 - TK_2 + 1 &= 0.70,\end{aligned}$$

which are easily solved for the coefficients and evaluated for $T = 0.1$ sec

$$K_1 = \frac{0.10}{T^2} = 10, \quad K_2 = \frac{0.35}{T} = 3.5.$$

The calculation of the gains using the method illustrated in the previous example becomes rather tedious when the order of the system (and therefore the order of the determinant to be evaluated) is greater than 2. A computer does not solve the tedium unless it is used to perform the algebraic manipulations necessary in expanding the determinant in Eq. (8.7) to obtain the characteristic equation. Therefore, other approaches have been developed to provide convenient computer-based solutions to this problem.

The algebra for finding the specific value of $\mathbf{K}$ is especially simple if the system matrices happen to be in the form associated with the block diagram of Fig. 4.8(c). This structure is called "control canonical form" because it is so useful in control law design. Referring to that figure and taking the state elements as the outputs of the delays (z^{-1} blocks), numbered from the left, we get (assuming $b_0 = 0$ for this case)

$$\Phi_c = \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \Gamma_c = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{H}_c = [b_1 \quad b_2 \quad b_3]. \quad (8.11)$$

Note that from Eq. (4.15), the characteristic polynomial of this system is $a(z) = z^3 + a_1z^2 + a_2z + a_3$. The key idea here is that the elements of the first row of Φ_c are exactly the coefficients of the characteristic polynomial of the system. If we now form the closed-loop system matrix $\Phi_c - \Gamma_c\mathbf{K}$, we find

$$\Phi_c - \Gamma_c\mathbf{K} = \begin{bmatrix} -a_1 - K_1 & -a_2 - K_2 & -a_3 - K_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \quad (8.12)$$

By inspection, we find that the characteristic equation of Eq. (8.12) is

$$z^3 + (a_1 + K_1)z^2 + (a_2 + K_2)z + (a_3 + K_3) = 0.$$

Thus, if the desired pole locations result in the characteristic equation

$$z^3 + \alpha_1z^2 + \alpha_2z + \alpha_3 = 0,$$

then the necessary values for control gains are

$$K_1 = \alpha_1 - a_1, \quad K_2 = \alpha_2 - a_2, \quad K_3 = \alpha_3 - a_3. \quad (8.13)$$

Conceptually, then, we have the canonical-form design method: Given an arbitrary (Φ, Γ) and a desired characteristic equation $\alpha(z) = 0$, we convert (by redefinition of the state) (Φ, Γ) to control form (Φ_c, Γ_c) and solve for the gain by Eq. (8.13). Because this gain is for the state in the control form, we must, finally, express the result back in terms of the original state elements. This method is sometimes used by CACSD packages because of the numerical advantages; however, the transformation is transparent to the designer, who generally prefers to use a state definition that is related to the physical system's characteristics.

8.1.2 Controllability

The first question this process raises is existence: Is it always possible to find an equivalent (Φ_c, Γ_c) for arbitrary (Φ, Γ) ? The answer is almost always "yes." The exception occurs in certain pathological systems, dubbed "uncontrollable," for which no control will give arbitrary pole locations. These systems have certain modes or subsystems that are unaffected by the control. Uncontrollability is best exhibited by a realization (selection of state elements) where each state element represents a natural mode of the system. If all the roots of the open-loop characteristic equation

$$|z\mathbf{I} - \Phi| = 0$$

are distinct, then Eq. (8.3) written in this way (normal mode or "Jordan canonical form") becomes

$$\mathbf{x}(k+1) = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} \Gamma_1 \\ \Gamma_2 \\ \vdots \\ \Gamma_n \end{bmatrix} u(k), \quad (8.14)$$

and explicitly exhibits the criterion for controllability: No element in Γ can be zero. If any Γ element was zero, the control would not influence that normal mode, and the associated state would remain uncontrolled. A good physical understanding of the system being controlled usually prevents any attempt to design a controller for an uncontrollable system; however, there is a mathematical test for controllability applicable to any system description, which may be an additional aid in discovering this condition; a discussion of this test is contained in Section 8.7.

8.1.3 Pole Placement Using CACSD

Ackermann's formula

MATLAB has two functions that perform the calculation of $\mathbf{K}$: `place.m` and `acker.m`. `Acker` is based on Ackermann's formula [Ackermann (1972)] and is satisfactory for SISO systems of order less than 10 and can handle systems with repeated roots. The relation is

$$\mathbf{K} = [0 \ \cdots \ 1] [\Gamma \ \Phi\Gamma \ \Phi^2\Gamma \ \cdots \ \Phi^{n-1}\Gamma]^{-1} \alpha_c(\Phi) \quad (8.15)$$

controllability matrix

where $\mathcal{C} = [\Gamma \ \Phi\Gamma \ \cdots]$ is called the **controllability matrix**, n is the order of the system or number of state elements, and we substitute Φ for z in $\alpha_c(z)$ to form

$$\alpha_c(\Phi) = \Phi^n + \alpha_1 \Phi^{n-1} + \alpha_2 \Phi^{n-2} + \cdots + \alpha_n \mathbf{I}, \quad (8.16)$$

where the α_i 's are the coefficients of the desired characteristic equation, that is,

$$\alpha_c(z) = |z\mathbf{I} - \Phi + \Gamma\mathbf{K}| = z^n + \alpha_1 z^{n-1} + \cdots + \alpha_n. \quad (8.17)$$

The controllability matrix, $\mathcal{C}$, must be full rank for the matrix to be invertible and for the system to be controllable.

`place` [Kautsky, Nichols, and Van Dooren (1985)] is best for higher order systems and can handle MIMO systems, but will not handle systems where the desired roots are repeated.

Note that these functions are used for both the continuous and discrete cases because they solve the same mathematical problems given by Eqs. (2.37) and (8.7). The only difference is that the desired root locations are in substantially different locations for the s -plane and z -plane and that $\mathbf{F}$, $\mathbf{G}$ have been replaced by Φ and Γ .

◆ Example 8.2 Control Law Pole Placement with MATLAB

Design a control law for the satellite attitude-control system as in Example 8.1. Place the z -plane closed-loop poles at $z = 0.8 \pm j0.25$.

Solution. The MATLAB statements

```
T = .1
```

```
Phi = [1 T; 0 1]
```

```
Gam = [T^2/2; T]
```

```
p = [.8+i*.25; .8-i*.25]
```

```
K = acker(Phi,Gam,p)
```

result in $\mathbf{K} = [10.25 \quad 3.4875]$. The difference between these values and those shown in Example 8.1 are due to round-off error in the hand calculation. place would have given the same answer.

A more complex system demonstrates the kind of difficulty you might encounter in using the pole placement approach. The specific difficulty is brought about by the necessity to pick n desired pole locations. Where should the higher frequency poles be picked? The system specifications typically help the designer pick only two of the desired poles. As discussed in Section 2.6, it is helpful to move poles as little as possible in order to minimize the required amount of control effort and to avoid exciting the system any more than necessary. The following example specifically illustrates how a poor choice of desired poles can cause an undesirable response, and how a wise choice of desired poles can drastically improve the situation.

◆ Example 8.3 Pole Placement for a 4th-Order System with MATLAB

Design a control law for the double mass-spring system in Appendix A.4 using d as the measurement. This system is representative of many systems where there is some flexibility between the measured output and control input. Assume the resonant mode has a frequency $\omega_n = 1$ rad/sec and damping $\zeta = 0.02$ and select a 10:1 ratio of the two masses. The parameters that provide these characteristics are: $M = 1$ kg, $m = 0.1$ kg, $b = 0.0036$ N-sec/m, and $k = 0.091$ N/m. Pick the sample rate to be 15 times faster than the resonance and show the free response to an initial condition of $d = 1$ m for two cases:

- (a) Pick all the poles at $z = 0.9$.
- (b) Pick the poles at $z = 0.9 \pm j0.05, 0.8 \pm j0.4$.

Discuss why the response is better for case (b).

Solution. From Eq. (A.17) we can write the state-space description as

$$\begin{aligned} \mathbf{x} &= [d \quad \dot{d} \quad y \quad \dot{y}]^T & \mathbf{H} &= [1 \quad 0 \quad 0 \quad 0] \\ \mathbf{F} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -0.91 & -0.036 & 0.91 & 0.036 \\ 0 & 0 & 0 & 1 \\ 0.091 & 0.0036 & -0.091 & -0.0036 \end{bmatrix} & \mathbf{G} &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad J = 0. \end{aligned}$$

The sample rate should be 15 rad/sec which translates to approximately $T = 0.4$ secs.

- (a) All poles at $z = 0.9$, the MATLAB script

$$\text{sysC} = \text{ss}(\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{J})$$

```
sysD = c2d(sysC,T,'zoh')
```

```
p = [.9;.9;.9;.9]
```

```
[phi,gam,H,J] = ssdata(sysD)
```

```
K=acker(phi,gam,p)
```

results in the feedback gain

$$\mathbf{K} = [0.650 \quad -0.651 \quad -0.645 \quad 0.718]. \quad (8.18)$$

place cannot be used for case (a) because of the repeated roots. For the response to an initial condition, the script

```
sysCL = feedback(K*sysD,1)
```

```
Xo = [1;0;0;0]
```

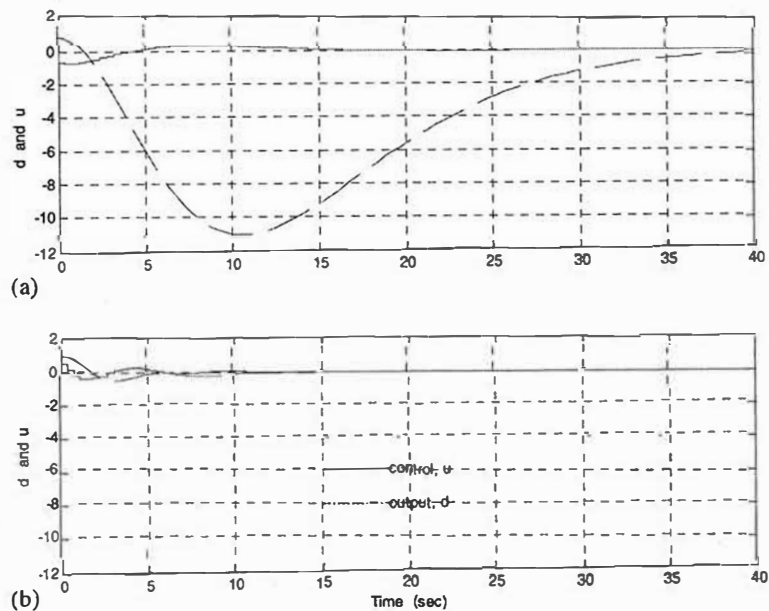
```
d = initial(sysCL,Xo)
```

produces the closed-loop response for $d(0) = 1$ m, shown in Fig. 8.2(a). It exhibits a response that is *much* larger than that of the initial condition, but the time characteristics are consistent with the selected poles.

(b) For the desired poles at $z = 0.9 \pm j0.05$, $0.8 \pm j0.4$, we modify the script above with

```
p = [.9+i*.05;.9-i*.05;.8+i*.4;.8-i*.4]
```

Figure 8.2
Initial condition response
for Example 8.3;
(a) desired poles all at
 $z = 0.9$, and (b) desired
poles at $z =$
 $0.9 \pm j0.05$, $0.8 \pm j0.4$



and this results in the feedback gain

$$\mathbf{K} = [-0.458 \quad -0.249 \quad 0.568 \quad 0.968], \quad (8.19)$$

which produces the response to an initial condition, $d = 1$ m, shown in Fig. 8.2(b). It exhibits *much* less response of d with no increase in control effort, although the resonant mode oscillations did influence the response with damping consistent with the poles selected. The primary reason for the superior response is that the oscillatory mode was not changed substantially by the control. Two of the selected poles ($z = 0.8 \pm j0.4$) have a natural frequency $\omega_n = 1$ rad/sec with a damping $\zeta \cong 0.2$. Therefore, the control is not attempting to change the natural frequency of the resonant mode at all; rather, the only task for the control in modifying this pole is to increase its damping from $\zeta = 0.02$ to $\zeta \cong 0.2$. Since the mode remains lightly damped, its oscillations are still visible on the output. The selected poles at $z = 0.9 \pm j0.05$ affect the overall motion of the system and their placement is less critical. Generally, pole selections with a damping $\zeta \cong 0.7$ give a better balance between system response and control usage. The control is clearly much more effective with these desired pole locations.

controller pole selection

So we see that the mechanics of computing the control law are easy, once the desired pole locations are known. The trick is to pick a good set of poles! The designer should iterate between pole selections and some other system evaluation to determine when the design is complete. System evaluation might consist of an initial-condition time response as shown in the example, a step response, steady-state errors, gain and phase margins, or the entire frequency-response shape. Pole placement by itself leaves something to be desired. But it is useful as a design tool to be used in conjunction with the other design methods discussed in Chapter 7 or as a part of an optimal design process that will be discussed in Chapter 9.

8.2 Estimator Design

The control law designed in the last section assumed that all state elements were available for feedback. Typically, not all elements are measured; therefore, the missing portion of the state needs to be reconstructed for use in the control law. We will first discuss methods to obtain an estimate of the entire state given a measurement of one of the state elements. This will provide the missing elements as well as provide a smoothed value of the measurement, which is often contaminated with random errors or “noise.” There are two basic kinds of estimates of the state, $\mathbf{x}(k)$: We call it the **current** estimate, $\hat{\mathbf{x}}(k)$, if based on measurements $y(k)$ up to and including the k th instant; and we call it the **predictor** estimate, $\bar{\mathbf{x}}(k)$, if based on measurements up to $y(k - 1)$. The idea eventually will be to let $u = -\mathbf{K}\hat{\mathbf{x}}$ or $u = -\mathbf{K}\bar{\mathbf{x}}$, replacing the true state used in Eq. (8.5) by its estimate.

8.2.1 Prediction Estimators

One method of estimating the state vector which might come to mind is to construct a model of the plant dynamics,

$$\bar{\mathbf{x}}(k+1) = \Phi \bar{\mathbf{x}}(k) + \Gamma u(k). \quad (8.20)$$

We know Φ , Γ , and $u(k)$, and hence this estimator should work if we can obtain the correct $\mathbf{x}(0)$ and set $\bar{\mathbf{x}}(0)$ equal to it. Figure 8.3 depicts this “open-loop” estimator. If we define the error in the estimate as

$$\tilde{\mathbf{x}} = \mathbf{x} - \bar{\mathbf{x}}, \quad (8.21)$$

and substitute Eqs. (8.3) and (8.20) into Eq. (8.21), we find that the dynamics of the resulting system are described by the estimator-error equation

$$\tilde{\mathbf{x}}(k+1) = \Phi \tilde{\mathbf{x}}(k). \quad (8.22)$$

Thus, if the initial value of $\bar{\mathbf{x}}$ is off, the dynamics of the estimate error are those of the uncompensated plant, Φ . For a marginally stable or unstable plant, the error will never decrease from the initial value. For an asymptotically stable plant, an initial error will decrease only because the plant and estimate will both approach zero. Basically, the estimator is running open loop and not utilizing any continuing measurements of the system’s behavior, and we would expect that it would diverge from the truth. However, if we feed back the difference between the measured output and the estimated output and constantly correct the model with this error signal, the divergence should be minimized. The idea is to construct a feedback system around the open-loop estimator with the estimated output error as the feedback. This scheme is shown in Fig. 8.4; the equation for it is

$$\bar{\mathbf{x}}(k+1) = \Phi \bar{\mathbf{x}}(k) + \Gamma u(k) + \mathbf{L}_p [y(k) - H \bar{\mathbf{x}}(k)], \quad (8.23)$$

Figure 8.3
Open-loop estimator

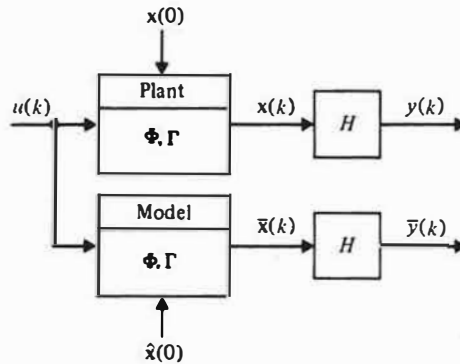
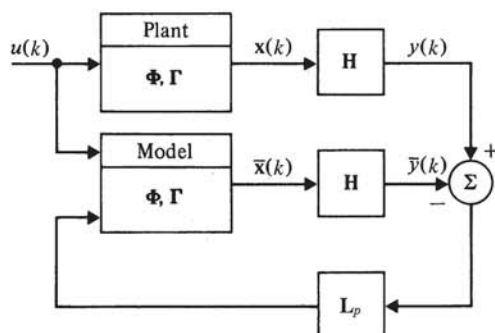


Figure 8.4
Closed-loop estimator



where $\mathbf{L}_p$ is the feedback gain matrix. We call this a **prediction estimator** because a measurement at time k results in an estimate of the state vector that is valid at time $k + 1$; that is, the estimate has been predicted one cycle in the future.

A difference equation describing the behavior of the estimation errors is obtained by subtracting Eq. (8.23) from Eq. (8.3). The result is

estimator error equation

$$\tilde{\mathbf{x}}(k+1) = [\Phi - \mathbf{L}_p \mathbf{H}] \tilde{\mathbf{x}}(k). \quad (8.24)$$

This is a homogeneous equation, but the dynamics are given by $[\Phi - \mathbf{L}_p \mathbf{H}]$; and if this system matrix represents an asymptotically stable system, $\tilde{\mathbf{x}}$ will converge to zero for any value of $\tilde{\mathbf{x}}(0)$. In other words, $\tilde{\mathbf{x}}(k)$ will converge toward $\mathbf{x}(k)$ regardless of the value of $\tilde{\mathbf{x}}(0)$ and could do so faster than the normal (open-loop) motion of $\mathbf{x}(k)$ if the estimator gain, $\mathbf{L}_p$, were large enough so that the roots of $\Phi - \mathbf{L}_p \mathbf{H}$ are sufficiently fast. In an actual implementation, $\tilde{\mathbf{x}}(k)$ will not equal $\mathbf{x}(k)$ because the model is not perfect, there are unmodelled disturbances, and the sensor has some errors and added noise. However, typically the sensed quantity and $\mathbf{L}_p$ can be chosen so that the system is stable and the error is acceptably small.

To find the value of $\mathbf{L}_p$, we take the same approach that we did when designing the control law. First, specify the desired estimator pole locations in the z -plane to obtain the desired estimator characteristic equation,

$$(z - \beta_1)(z - \beta_2) \cdots (z - \beta_n) = 0, \quad (8.25)$$

where the β 's are the desired estimator pole locations⁴ and represent how fast the estimator state vector converges toward the plant state vector. Then form the characteristic equation from the estimator-error equation (8.24),

$$|z\mathbf{I} - \Phi + \mathbf{L}_p \mathbf{H}| = 0. \quad (8.26)$$

⁴ The following sections discuss how one should select these poles in relation to the control poles and how both sets of poles appear in the combined system. The issue was also discussed for the continuous case in Section 2.6.2.

Equations (8.25) and (8.26) must be identical. Therefore, the coefficient of each power of z must be the same, and, just as in the control case, we obtain n equations in n unknown elements of $\mathbf{L}_p$ for an n th-order system.

◆ Example 8.4 Estimator Design for Satellite Attitude

Construct an estimator for the same case as in Example 8.1, where the measurement is the position state element, x_1 , so that $\mathbf{H} = [1 \ 0]$ as given by Eq. (4.47). Pick the desired poles of the estimator to be at $z = 0.4 \pm j0.4$ so that the s -plane poles have $\zeta \cong 0.6$ and ω_n is about three times faster than the selected control poles (see Fig. 8.1).

Solution. The desired characteristic equation is then (approximately)

$$z^2 - 0.8z + 0.32 = 0, \quad (8.27)$$

and the evaluation of Eq. (8.26) for any estimator gain $\mathbf{L}_p$ leads to

$$\left| z \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} L_{p1} \\ L_{p2} \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right| = 0$$

or

$$z^2 + (L_{p1} - 2)z + TL_{p2} + 1 - L_{p1} = 0. \quad (8.28)$$

Equating coefficients in Eqs. (8.27) and (8.28) with like powers of z , we obtain two simultaneous equations in the two unknown elements of $\mathbf{L}_p$

$$\begin{aligned} L_{p1} - 2 &= -0.8 \\ TL_{p2} + 1 - L_{p1} &= 0.32, \end{aligned}$$

which are easily solved for the coefficients and evaluated for $T = 0.1$ sec

$$L_{p1} = 1.2, \quad L_{p2} = \frac{0.52}{T} = 5.2. \quad (8.29)$$

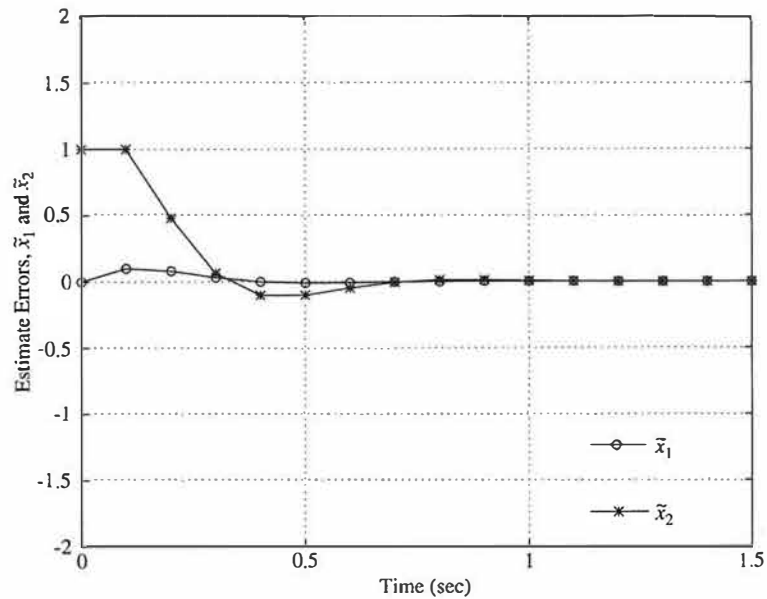
Thus the estimator algorithm would be Eq. (8.23) with $\mathbf{L}_p$ given by Eq. (8.29), and the equations to be coded in the computer are

$$\begin{aligned} \tilde{x}_1(k+1) &= \tilde{x}_1(k) + 0.1\tilde{x}_2(k) + 0.005u(k) + 1.2[y(k) - \tilde{x}_1(k)] \\ \tilde{x}_2(k+1) &= \tilde{x}_2(k) + 0.1u(k) + 5.2[y(k) - \tilde{x}_1(k)]. \end{aligned}$$

Figure 8.5 shows the time history of the estimator error from Eq. (8.24) for the gains in Eq. (8.29) and with an initial error of 0 for the position estimate and 1 rad/sec for the velocity estimate.

The transient settling in $\tilde{x}_2$ could be hastened by higher values of the gains, $\mathbf{L}_p$, that would result by selecting faster estimator poles, but this would occur at the expense of more response of both $\tilde{x}_1$ and $\tilde{x}_2$ to measurement noise.

Figure 8.5
Time history of the
prediction estimator
error



It is important to note that an initial estimator transient or, equivalently, the occurrence of an unmodelled input to the plant, can be a rare event. If the problem is one of regulation, the initial transient might be unimportant compared to the long-term performance of the estimator in the presence of noisy measurements. In the regulator case with very small plant disturbances, very slow poles (maybe even slower than the control poles) and their associated low estimator gains would give smaller estimate errors. Optimal selection of estimator gains based on the system's noise characteristics will be discussed in Chapter 9.

8.2.2 Observability

Given a desired set of estimator poles, is L_p uniquely determined? It is, provided y is a scalar and the system is "observable." We might have an unobservable system if some of its modes do not appear at the given measurement. For example, if only derivatives of certain states are measured and these states do not affect the dynamics, a constant of integration is obscured. This situation occurs with a $1/s^2$ plant if only velocity is measured, for then it is impossible to deduce the initial condition of the position. For an oscillator, a velocity measurement is sufficient to estimate position because the acceleration, and consequently the velocity, observed are affected by position. A system with cycle delays can also be unobservable because the state elements representing the delays have no influence on the measurement and can therefore not be reconstructed by the measurements. A mathematical test for observability is stated in the next section

as a necessary condition for the solution of Ackermann's formula; its proof is given in Section 8.7.

8.2.3 Pole Placement Using CACSD

If we take the transpose of the error-equation system matrix from Eq. (8.24)

$$[\Phi - L_p H]^T = \Phi^T - H^T L_p^T,$$

we see that the result is the same form as the system matrix $\Phi - \Gamma K$ of the control problem from Eq. (8.6), and the mathematics of the solution is the same. Therefore, to solve the problem, we substitute Φ^T for Φ , H^T for Γ and L_p^T for K , and use the control-design results. Making the substitutions in Eq. (8.15) results in Ackermann's estimator formula

observability matrix

$$L_p = \alpha_e(\Phi) \begin{bmatrix} H \\ H\Phi \\ H\Phi^2 \\ \vdots \\ H\Phi^{n-1} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad (8.30)$$

where

$$\alpha_e(\Phi) = \Phi^n + \alpha_1 \Phi^{n-1} + \alpha_2 \Phi^{n-2} + \cdots + \alpha_n I, \quad (8.31)$$

and the α_i 's are the coefficients of the desired characteristic equation, that is

$$\alpha_e(z) = (z - \beta_1)(z - \beta_2) \cdots (z - \beta_n) = z^n + \alpha_1 z^{n-1} + \cdots + \alpha_n. \quad (8.32)$$

The coefficient matrix with rows $H\Phi^j$ is called the **observability matrix** and must be full rank for the matrix to be invertible and for the system to be observable.

For calculation of L_p with MATLAB, either `acker` or `place` can be used by invoking the substitutions above. The same restrictions apply that existed for the control problem.

◆ Example 8.5 Predictor Estimator Pole Placement with MATLAB

Design an estimator for the satellite attitude-control system as in Example 8.4. Place the z -plane poles at $z = 0.4 \pm j0.4$.

Solution. The MATLAB statements

```
T = .1
Phi = [1 T; 0 1]
Gam = [T^2/2; T]
p = [.4+i*.4; .4-i*.4]
```

$$L_p = \text{acker}(\Phi', H', p)'$$

result in

$$L_p = \begin{bmatrix} 1.2 \\ 5.2 \end{bmatrix}.$$

place would have given the same answer.

8.2.4 Current Estimators

As was already noted, the previous form of the estimator equation (8.23) arrives at the state vector estimate $\bar{\mathbf{x}}$ after receiving measurements up through $y(k-1)$. This means that the current value of control⁵ does not depend on the most current value of the observation and thus might not be as accurate as it could be. For high-order systems controlled with a slow computer or any time the sample periods are comparable to the computation time, this delay between the observation instant and the validity time of the control output can be a blessing because it allows time for the computer to complete the calculations. In many systems, however, the computation time required to evaluate Eq. (8.23) is quite short compared to the sample period, and the delay of almost a cycle between the measurement and the proper time to apply the resulting control calculation represents an unnecessary waste. Therefore, it is useful to construct an alternative estimator formulation that provides a current estimate $\hat{\mathbf{x}}$ based on the current measurement $y(k)$.⁶ Modifying Eq. (8.23) to yield this feature, we obtain

$$\hat{\mathbf{x}}(k) = \bar{\mathbf{x}}(k) + \mathbf{L}_e[y(k) - \mathbf{H}\bar{\mathbf{x}}(k)], \quad (8.33)$$

where $\bar{\mathbf{x}}(k)$ is the predicted estimate based on a model prediction from the previous time estimate, that is

$$\bar{\mathbf{x}}(k) = \Phi\hat{\mathbf{x}}(k-1) + \Gamma u(k-1). \quad (8.34)$$

Control from this estimator cannot be implemented exactly because it is impossible to sample, perform calculations, and output with absolutely no time elapsed. However, the calculation of $u(k)$ based on Eq. (8.33) can be arranged to minimize computational delays by performing all calculations before the sample instant that do not directly depend on the $y(k)$ measurement. In Chapter 3, we also had a $u(k)$ that was dependent on $y(k)$ (see Table 3.1); and here, too, we organized the calculations to minimize the delay. If this latency in the implementation causes significant errors in the performance of the system compared to the analysis,

⁵ We plan to use $u(k) = -\mathbf{K}\bar{\mathbf{x}}$ in place of $u(k) = -\mathbf{K}\mathbf{x}$.

⁶ This form of the equations is used in the Kalman filter, which is discussed in Chapter 9.

it could be modeled and accounted for in the estimator equations by using the results of Section 4.3.4.

To help understand the difference between the prediction and current form of estimation—that is, Equations (8.23) and (8.33)—it is useful to substitute Eq. (8.33) into Eq. (8.34). This results in

$$\tilde{\mathbf{x}}(k+1) = \Phi\tilde{\mathbf{x}}(k) + \Gamma u(k) + \Phi\mathbf{L}_c[y(k) - \mathbf{H}\tilde{\mathbf{x}}(k)]. \quad (8.35)$$

Furthermore, the estimation-error equation for $\tilde{\mathbf{x}}(k)$, obtained by subtracting Eq. (8.3) from Eq. (8.35), is

$$\tilde{\mathbf{x}}(k+1) = [\Phi - \Phi\mathbf{L}_c\mathbf{H}]\tilde{\mathbf{x}}(k). \quad (8.36)$$

By comparing Eqs. (8.35) with (8.23) and (8.36) with (8.24), we can conclude that $\tilde{\mathbf{x}}$ in the current estimator equation, (8.33), is the same quantity as $\tilde{\mathbf{x}}$ in the predictor estimator equation, (8.23), and that the estimator gain matrices are related by

$$\mathbf{L}_p = \Phi\mathbf{L}_c. \quad (8.37)$$

The relationship between the two estimates is further illuminated by writing Eqs. (8.33) and (8.34) as a block diagram, as in Fig. 8.6. It shows that $\hat{\mathbf{x}}$ and $\tilde{\mathbf{x}}$ represent different outputs of the same estimator system.

We can also determine the estimator-error equation for $\hat{\mathbf{x}}$ by subtracting Eq. (8.33) from (8.3). The result is⁷

$$\tilde{\mathbf{x}}(k+1) = [\Phi - \mathbf{L}_c\mathbf{H}\Phi]\tilde{\mathbf{x}}(k). \quad (8.38)$$

The two error equations, (8.36) and (8.38), can be shown to have the same roots, as should be the case because they simply represent the dynamics of different outputs of the same system. Therefore, we could use either form as the basis for computing the estimator gain, $\mathbf{L}_c$. Using Eq. (8.38), we note that it is similar to Eq. (8.24) except that $\mathbf{H}\Phi$ appears instead of $\mathbf{H}$. To use Ackermann's formula for $\mathbf{L}_c$, we use Eq. (8.30) with $\mathbf{H}$ replaced by $\mathbf{H}\Phi$ and find

$$\mathbf{L}_c = \alpha_e(\Phi) \begin{bmatrix} \mathbf{H}\Phi \\ \mathbf{H}\Phi^2 \\ \mathbf{H}\Phi^3 \\ \vdots \\ \mathbf{H}\Phi^n \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}, \quad (8.39)$$

where $\alpha_e(\Phi)$ is based on the desired root locations and is given by Eqs. (8.31) and (8.32). To use the control form of Ackermann's formula to perform the calculations, we take the transpose of $\Phi - \mathbf{L}_c\mathbf{H}\Phi$ and get $\Phi^T - \Phi^T\mathbf{H}^T\mathbf{L}_c^T$, which is the same form as the system matrix $\Phi - \Gamma\mathbf{K}$ of the control problem. Therefore

⁷ Equation (8.21) defines $\tilde{\mathbf{x}}$ to be $\mathbf{x} - \hat{\mathbf{x}}$; however, we use $\tilde{\mathbf{x}}$ here to be $\mathbf{x} - \hat{\mathbf{x}}$. In both cases, $\tilde{\mathbf{x}}$ refers to the estimator error.